

5.2) 6) $\text{supp } a, b \in \mathbb{R}, a \neq 0$

$$f: \mathbb{R} \rightarrow \mathbb{R}, f(x) = ax + b$$

One to One:

$$\text{supp } x, y \in \mathbb{R} \text{ \& } f(x) = f(y) \longrightarrow x = y$$

$$\text{then } ax + b = ay + b \iff ax = ay \iff x = y$$

Onto: $\text{supp } y \in \mathbb{R}$

$$\exists x \in \mathbb{R} (f(x) = y)$$

$$\text{let } x = \frac{y-b}{a}, \text{ then}$$

$$ax + b = y$$

$$f(x) = a\left(\frac{y-b}{a}\right) + b = y - b + b = y$$

$$x = \frac{y-b}{a}$$

5.2) 7) $f: \mathbb{R}^+ \rightarrow \mathbb{R}, f(x) = \frac{1}{x} - x$

(a) One to One:

$$\text{supp } x, y \in \mathbb{R}^+ \text{ \& } f(x) = f(y) \longrightarrow x = y$$

$$\text{then } \frac{1}{x} - x = \frac{1}{y} - y \quad \text{--- (1)}$$

Proof By Contradiction: $\text{supp } x \neq y$

Case 1) $x > y$.

$$\text{then } \frac{1}{x} < \frac{1}{y} \iff \frac{1}{x} - x < \frac{1}{y} - x < \frac{1}{y} - y \quad (\because x > y)$$

but this is contradiction to (1).

Case 2) $x < y$

$$\text{then } \frac{1}{x} > \frac{1}{y} \iff \frac{1}{x} - x > \frac{1}{y} - x > \frac{1}{y} - y \text{ which is contradiction to (1) again.}$$

(b) Onto:

$$\text{supp } y \in \mathbb{R} \longrightarrow \exists x \in \mathbb{R}^+ (f(x) = y)$$

If we put $f(x) = y$, then

$$\frac{1}{x} - x = y \iff 1 - x^2 = xy \iff x^2 + xy - 1 = 0.$$

By Quadratic formula, this yields $x = \frac{-y \pm \sqrt{y^2 + 4}}{2}$ & $y^2 + 4 > 0$

clearly, so we can consider 2 cases. 2

Case 1) $x = \frac{-y + \sqrt{y^2 + 4}}{2}$

Then

$$\begin{aligned}
 f(x) &= \frac{1}{\frac{-y + \sqrt{y^2 + 4}}{2}} - \frac{-y + \sqrt{y^2 + 4}}{2} \\
 &= \frac{2}{-y + \sqrt{y^2 + 4}} - \frac{-y + \sqrt{y^2 + 4}}{2} \\
 &= \frac{4 - (-y + \sqrt{y^2 + 4})^2}{2(-y + \sqrt{y^2 + 4})} = \frac{4 - (y^2 + y^2 + 4 + 2(-y)\sqrt{y^2 + 4})}{2(-y + \sqrt{y^2 + 4})} \\
 &= \frac{4 - 2y^2 - 4 - 2y\sqrt{y^2 + 4}}{2(-y + \sqrt{y^2 + 4})} = \frac{-2(y^2 + y\sqrt{y^2 + 4})}{2(-y + \sqrt{y^2 + 4})} \\
 &= \frac{-y^2 - y\sqrt{y^2 + 4}}{-y + \sqrt{y^2 + 4}} = \frac{y(-y + \sqrt{y^2 + 4})}{-y + \sqrt{y^2 + 4}} = y
 \end{aligned}$$

Case 2) $x = \frac{-y - \sqrt{y^2 + 4}}{2}$

This is similar to case 1.

(c) $g: \mathbb{R}^+ \rightarrow \mathbb{R}$, $g(x) = \frac{1}{x} + x$

One to One?

g is not one to one as x & $\frac{1}{x}$ both give same output

e.g. $\frac{1}{2} + 2 = \frac{1}{0.5} + 0.5$ but $2 \neq 0.5$

Onto?

g is not onto. It's clear to see if $y < 0$ then $x = \frac{1}{x}$ cannot be y as $x \in \mathbb{R}^+$. Formally we can show that $\exists y \in \mathbb{R} \forall x \in \mathbb{R}^+ (f(x) \neq y)$

Supp $y = -1$ (we get this after calculating x by Quadratic Formula)

Supp $x \in \mathbb{R}^+$ & supp $f(x) = y$ $\therefore$ Proof By Contradiction

then $\frac{1}{x} + x = y$. By Quadratic formula, this gives

$$x = \frac{y \pm \sqrt{y^2 - 4}}{2} \quad \text{if } y = -1 \text{ then } \sqrt{(-1)^2 - 4} = \sqrt{-3} \notin \mathbb{R}$$

so there is no real x that satisfies $f(x) = -1$.